Integrating Suricata with Wazuh

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Introduction

Intrusion Detection Systems (IDS) and Intrusion Prevention Systems (IPS) are essential components in a layered cybersecurity strategy. These systems inspect network traffic for known and unknown threats to help detect and mitigate attacks.

1 IDS vs. IPS Overview

- **IDS (Intrusion Detection System):**
  - Detects threats by inspecting traffic using:
    - Signatures (rules)
    - IoCs (hashes, domains, URLs, TLS/SSH fingerprints)
    - Lua scripting
  - Generates alerts and logs but does not block traffic.
- **IPS (Intrusion Prevention System):**
  - Takes proactive action by dropping or allowing traffic based on inspection results.

2 About Suricata

Suricata is an open-source network threat detection engine capable of functioning as both an IDS and an IPS. It is widely used in enterprise and government environments.

- By default, Suricata runs in IDS mode and logs suspicious traffic.
- In IPS mode, it can block malicious packets in real time.

3 Setting Up Suricata on Windows

3.1 Download and Install

1. Visit: https://suricata.io/download/
2. Download the Windows installer (.msi)
3. Run the installer with default settings

Suricata installs to:

```
C:\Program Files\Suricata\
```

Contents include:

- suricata.exe – the main executable
- suricata.yaml – configuration file
- rules\ – rule files directory
- log\ – log output folder

3.2 Install and Verify Npcap

Suricata requires Npcap to capture network traffic.

Step 1: Download from https://npcap.com/#download

During installation:

- Enable WinPcap API-compatible mode
- Enable startup at boot

Step 2: Verify that Npcap is running:

```
Get-Service -Name npcap
```

Expected output:

```
Status Name DisplayName
------ ---- -----------
Running npcap Npcap Packet Driver (NPCAP)
```

If not running:

```
Start-Service -Name npcap
```

4 Configure Suricata

Open Configuration

```
C:\Program Files\Suricata\suricata.yaml
```

Define Network Interface

To view interfaces:

```
Get-NetAdapter | Select Name, Status
```

Example config snippet:

```
af-packet:
- interface: Ethernet
```

Add Rule Files

In suricata.yaml, locate and update:

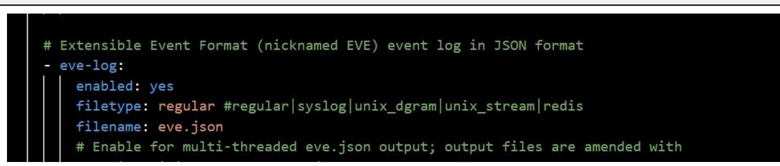
```
rule-files:
-   local.rules
-   emerging-all.rules
```

Make sure emerging-all.rules exists in the rules\ folder.

Enable JSON Logging

Enable EVE JSON output:

```
outputs:
  - eve-log:
      enabled: yes filetype:
      regular filename: eve.json
```



```
# Extensible Event Format (nicknamed EVE) event log in JSON format
- eve-log:
    enabled: yes
    filetype: regular #regular|syslog|unix_dgram|unix_stream|redis
    filename: eve.json
    # Enable for multi-threaded eve.json output; output files are amended with
```

Figure 1: Figure 1: JSON logging configuration in suricata.yaml

5 Running Suricata

In PowerShell (Admin):

```
cd "C:\Program Files\Suricata\" suricata -c
suricata.yaml -i <interface>
```

Replace <interface> with your adapter name.

Suricata Log Files

```
C:\Program Files\Suricata\log\
```

Important files:

- eve.json – JSON alerts
- fast.log – quick alerts
- stats.log – system stats

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Step 1: Confirm EVE JSON Output

Ensure Suricata is writing logs to:

```
C:\Program Files\Suricata\log\eve.json
```

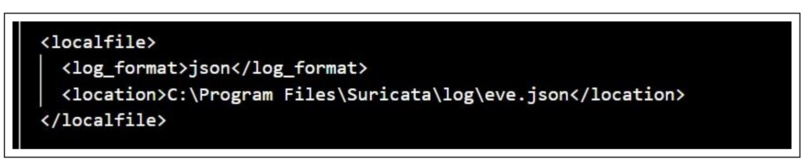
Step 2: Configure Wazuh Agent

Edit:

```
C:\Program Files (x86)\ossec-agent\ossec.conf
```

Add:

```
<localfile>
<log_format>json</log_format>
<location>C:\Program Files\Suricata\log\eve.json</location>
</localfile>
```



```
<localfile>
  <log_format>json</log_format>
  <location>C:\Program Files\Suricata\log\eve.json</location>
</localfile>
```

Figure 2: Figure 2: Wazuh Agent config for EVE JSON

Step 3: Restart the Wazuh Agent

```
Restart-Service -Name wazuh
```

Step 4: Generate Alerts

Use test traffic such as:

```
ping google.com
```

Step 5: View Alerts in Wazuh

1. Open Wazuh Dashboard
2. Navigate to **Threat Hunting**
3. Filter by Suricata

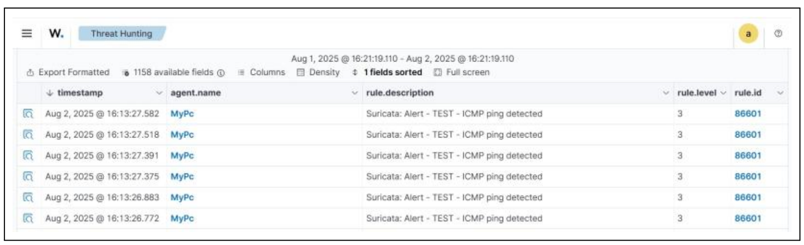


Figure 3: Figure 3: Suricata alert in Wazuh dashboard

7 Conclusion

Suricata is a powerful, flexible IDS/IPS that integrates easily with Wazuh for centralized visibility and alerting. With the proper configuration, your Windows environment gains advanced network monitoring capabilities with open-source tools.